Northeastern University

Report for Experiment #16

Electric Field and Electric Potential

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Abstract

The aim of this experiment was to investigate the relationship between electric potential and electric field using parallel plate and concentric circular electrodes. A 10 V DC supply was applied, and voltages were measured with a digital multimeter to map equipotential lines. For the parallel plate configuration, the measured average electric field was –0.91 ± 0.04 V/cm, compared to the theoretical value of –1.00 V/cm, yielding a percent error of 1.7%. For the concentric electrodes, the potential decreased logarithmically with radius. At r = 2.05 ± 0.11 cm, the measured potential was 7.00 V, while the theoretical prediction was 7.42 V. The corresponding electric field at this radius was 0.71 ± 0.07 V/cm, compared to the theoretical value of 0.78 V/cm. Across all radii, experimental values followed the expected logarithmic and inverse relations, with deviations primarily due to electrode contact resistance, marking accuracy, and finite-size effects. Overall, the results confirmed that equipotential lines are perpendicular to electric field lines and that measured fields agreed with theoretical predictions within experimental uncertainty.

Introduction

In this experiment, we investigated how electric potential and electric field are related by measuring potential between two different electrode configurations. Electric potential is a scalar quantity that describes the potential energy per unit charge at a point, while the electric field is a vector quantity that describes the force per unit charge and has both magnitude and direction. A test charge q placed in an electric field E experience a force given by

$$F = qE \quad (16.1)$$

To study the potential, two electrodes can be placed on a conducting surface and connected to a voltage source. Assigning one electrode as the zero-potential reference allows us to measure the potential at other points. Repeating this process creates equipotential lines curves along which the potential remains constant. Since the electric field is the negative gradient of the potential, we can calculate its x-component between two points a and b as

$$E_x = -\frac{\Delta V}{\Delta x} = -\frac{V_b - V_a}{x_b - x_a} \quad (16.2)$$

More generally, the change in potential between two points is related to the electric field and displacement by

$$\Delta V = -E \cdot \Delta r \quad (16.3)$$

Integrating over a path gives the potential difference between two points:

$$V_b - V_a = -\int_a^b E \, dr \quad (16.4)$$

Visually, equipotential lines are always drawn perpendicular to electric field lines. The density of field lines represents field strength: closely spaced lines indicate stronger fields, while widely spaced lines correspond to weaker fields.

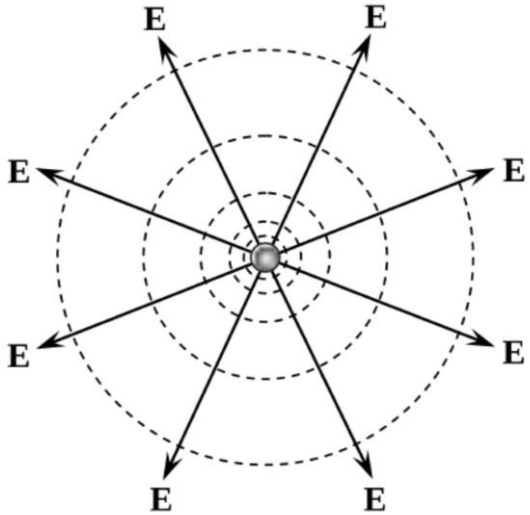


Figure 1 – Equipotential lines (dashed) and electric field lines (solid) around a point charge.

In this lab, we applied a 10 V potential difference to two electrode arrangements: parallel plates and concentric circles. For the parallel plates, we expected a nearly uniform electric field where potential varies linearly with distance. For the concentric setup, we expected a radial electric field decreasing with distance, and a potential that follows a logarithmic function of radius. By mapping equipotential lines with a voltmeter, we were able to calculate the electric field, compare it to theoretical predictions, and examine how geometry affects field distribution.

Investigation 1

For the first investigation, we examined the potential and electric field between two parallel electrodes separated by 10 cm. While the plates are not infinitely long, Gauss' law predicts that near the center, the field should approximate that of infinite plates. The theoretical expression for the field is constant:

$$E = \frac{\Delta V}{d}$$

with direction perpendicular to the plates. Since V (x = 0) = 0 at the grounded electrode, the potential is given by

$$V(x) = Ex$$

where d = 0.10 m and ΔV = 10 V. This predicts a uniform field of

$$E_{theo} = \frac{10}{0.10} = -100\ V/m$$

We placed a sheet of conductive paper on the rubber pad and positioned the electrodes 10 cm apart. Using a grease pencil, we outlined the electrode boundaries. The left electrode was grounded, and the right electrode was connected to the +10 V supply.

The voltmeter was connected with its ground lead to the negative electrode and the probe tip to measure potential across the paper. The scale was set to read up to 10 V. We traced equipotential lines at 2 V, 3 V, 4 V, 5 V, 6 V, 7 V, and 8 V, marking their positions. To minimize contact issues, one partner held the paper flat while the other measured.

At each equipotential line, we recorded positions x_a and x_b from the negative electrode. The midpoint gave the average position:

$$x_{avg} = \frac{x_a + x_b}{2}$$

The electric field was then calculated from the potential difference:

$$E = -\frac{\Delta V}{\Delta x}$$

Uncertainty in x_{avg} was taken as half the pencil line thickness, δx_{avg} = 0.035 cm.

The raw and processed values are shown in Table 1.

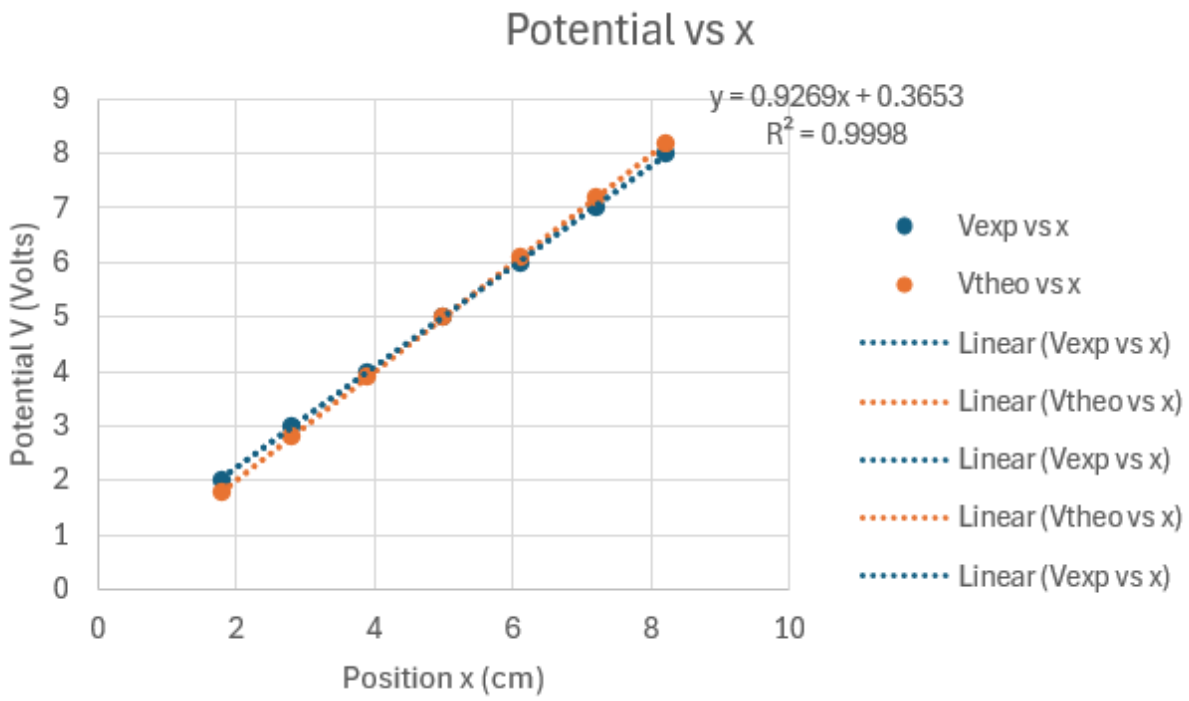
xa	δxa	xb	δxb	Va	Vb	δV	xavg	δxavg	Eexp	δE/E	Etheo	Vtheo
0	0.05	1.8	0.05	0	2	0.02	0.9	0.035355	-1.11111	0.029523	-1	1.8
1.8	0.05	2.8	0.05	2	3	0.03	2.3	0.035355	-1	0.020466	-1	2.8
2.8	0.05	3.9	0.05	3	4	0.04	3.35	0.035355	-0.90909	0.016259	-1	3.9
3.9	0.05	5	0.05	4	5	0.05	4.45	0.035355	-0.90909	0.014142	-1	5
5	0.05	6.1	0.05	5	6	0.06	5.55	0.035355	-0.90909	0.01293	-1	6.1
6.1	0.05	7.2	0.05	6	7	0.07	6.65	0.035355	-0.90909	0.012175	-1	7.2
7.2	0.05	8.2	0.05	7	8	0.08	7.7	0.035355	-1	0.011712	-1	8.2

Table 1 – Processed and Raw Data for Parallel Electrodes

Voltage vs Average Position

A plot of V vs x_{avg} produced a straight line with positive slope.

This agrees with the theoretical prediction V(x) = Ex.

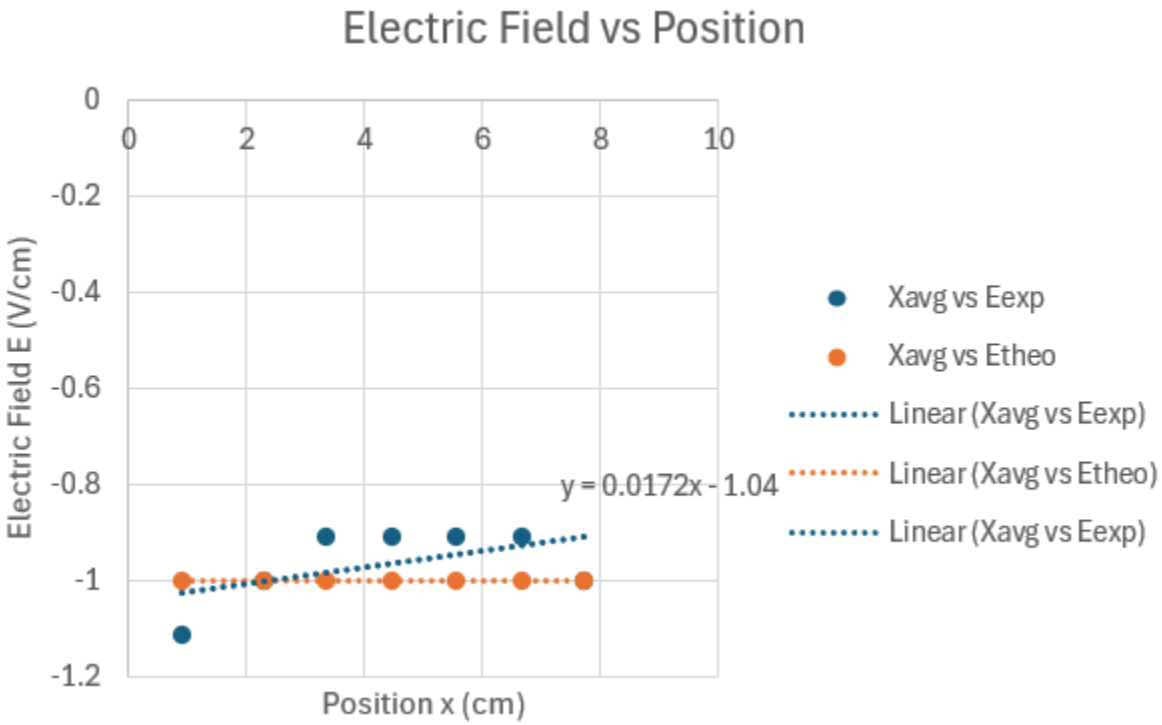


Graph 1 – Voltage vs Average Position with Theoretical Line

Electric Field vs Average Position

A plot of E vs x_{avg} showed the measured field values clustered near the theoretical constant value of – 100 V/m.

Some scatter is present due to measurement uncertainty and electrode contact imperfections.



Graph 2 – Electric Field vs Average Position with Theoretical Field

From Graph 1, the measured potential increased linearly with distance from the grounded electrode, consistent with theory. For example, at x_{avg} = 2.30 cm, the measured voltage was 3 V, close to the theoretical value of 2.8 V.

From Graph 2, the experimental electric field values fluctuated around − 0.91 to − 1.11 V/cm, compared to the theoretical − 1.0 V/cm. The relative error δ E / E ranged from 0.0295 to 0.0117, indicating good agreement within uncertainty.

Slight deviations were observed near the edges of the plates, consistent with fringe field effects, were equipment lines curve outward. This caused some measured values to drift above or below the theoretical constant.

Investigation 2

In the second part of the experiment, we examined the electric potential and electric field distribution between concentric cylindrical electrodes. This setup is analogous to the cross-section of a coaxial cable, where the outer conductor is grounded, and the inner conductor is held at a positive potential. According to Gauss's Law, the electric field in such a system is radial and decreases inversely with distance from the center, while the potential follows a logarithmic dependence.

The theoretical expressions are:

$$E(r) = \frac{V_0}{\ln(\frac{b}{a})} * \frac{1}{r}$$

$$V(r) = V_0 * \frac{\ln(b) - \ln(r)}{\ln(b) - \ln(a)}$$

where:

- V_0 = 10.0 V is the applied voltage,
- a = 1.15 cm is the inner electrode radius,
- b = 10.8 cm is the outer electrode radius,
- r is the radial distance from the center.

The same sheet of conducting paper used in Investigation 1 was reused here, with a separate section reserved for the concentric setup. A coordinate system was drawn at the center, and the inner brass cylinder was placed at the origin, surrounded by the larger outer ring electrode. Care was taken to align the electrodes concentrically to minimize systematic error.

The outer electrode was grounded, and the inner electrode was connected to the +10 V terminal of the power supply. The voltmeter was grounded to the outer electrode, while the probe was used to record potential values at various points on the conducting paper. Equipotential points corresponding to 2 V, 3 V, 4 V, 5 V, 6 V, 7 V, and 8 V were measured at angular intervals of 45 degrees. These points were then connected to form equipotential curves, which were observed to be nearly circular.

For each equipment, four radial measurements were taken (along the x, y, and diagonal directions). The average radius was calculated as:

$$r_{avg} = \frac{r_1 + r_2 + r_3 + r_4}{4}$$

The uncertainty in r_{avg} was determined from:

$$\delta r = \frac{\sigma}{\sqrt{N}}$$

where σ is the standard deviation of the four measurements, and N = 4.

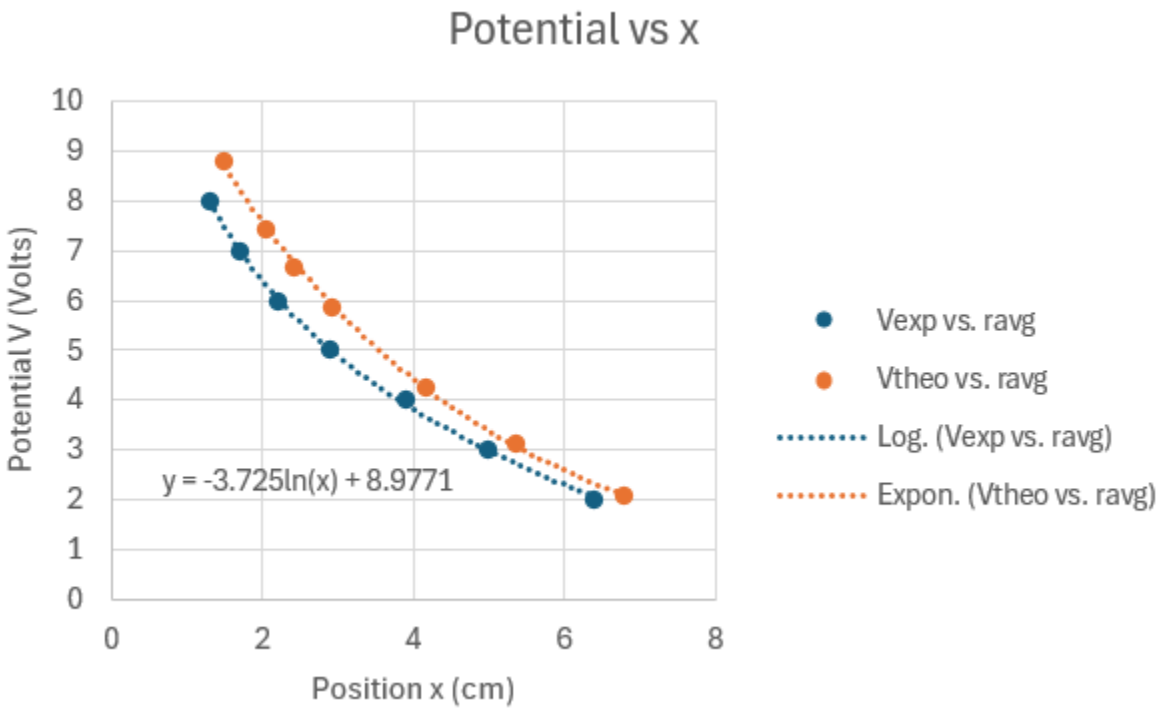
The processed and raw data for concentric electrodes is summarized below:

r1	r2	r3	r4	ravg	δravg	Vexp	Vtheo	α	β	Vα	Vβ	rE	Eexp	Etheo	1/rE
1.4	1.6	1.5	1.5	1.5	0.070711	2	8.813711	10.8	6.4	0	2	8.6	0.454545	0.519153	0.116279
2.2	2	2.1	1.9	2.05	0.111803	3	7.419047	6.4	5	2	3	5.7	0.714286	0.783283	0.175439
2.6	2.4	2.5	2.2	2.425	0.147902	4	6.669011	5	3.9	3	4	4.45	0.909091	1.003307	0.224719
2.9	2.86	2.9	3	2.915	0.05172	5	5.847333	3.9	2.9	4	5	3.4	1	1.313152	0.294118
4.3	4.1	4.3	4	4.175	0.129904	6	4.243408	2.9	2.2	5	6	2.55	1.428571	1.750869	0.392157
5.5	5.2	5.5	5.2	5.35	0.15	7	3.136238	2.2	1.7	6	7	1.95	2	2.289598	0.512821
6.7	6.6	7	6.9	6.8	0.158114	8	2.065483	1.7	1.3	7	8	1.5	2.5	2.976477	0.666667

Table 2 – Processed and raw data for concentric electrodes

This table includes the measured radii, average radius, uncertainty, experimental voltages, theoretical voltages (Eq. 16.6), and both experimental and theoretical electric field values (Eq. 16.7).

The potential distribution was first analyzed by plotting experimental voltage as a function of the average radius. Theoretical values were calculated from Equation (16.6) and overlaid on the same plot.



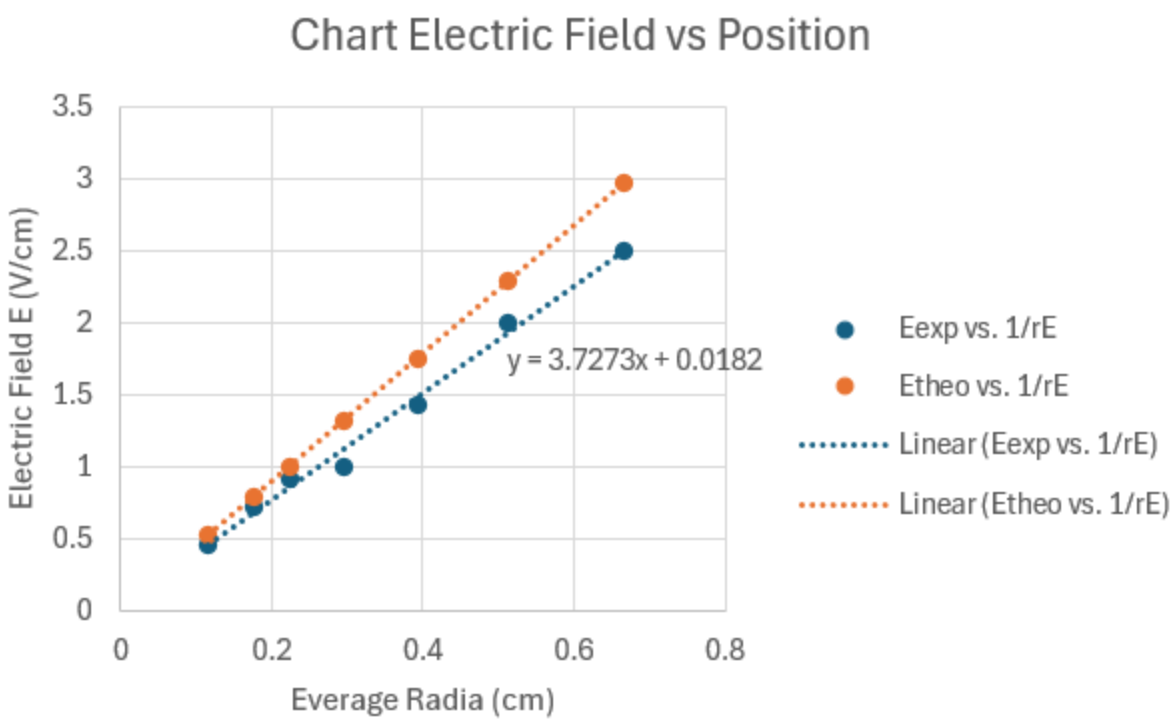
Graph 3 – Voltage vs average radius (with logarithmic fit and theoretical curve)

As expected, the experimental data followed a logarithmic decay, with larger voltage drops close to the inner electrode and more gradual decreases toward the outer ring. The logarithmic fit matched the experimental data within the uncertainties, though small deviations occurred at the 2 V and 7 V equipotential, which may be due to measurement error in locating the equipotential curves.

The electric field was then calculated from the potential differences between neighboring equipotential lines:

$$E = -\frac{\Delta V}{\Delta r}$$

These values were plotted against 1/r, since theory predicts a linear relationship with slope proportional to $\frac{V_0}{\ln(\frac{b}{a})}$



Graph 4 – Electric field vs 1/r (with linear fit and theoretical comparison)

The resulting plot demonstrated that the experimental data aligned well with the expected linear trend. The slope of the best-fit line was slightly higher than the theoretical prediction, suggesting that the measured electric field was stronger than expected. This discrepancy could be attributed to small electrode misalignments, finite electrode thickness, or edge effects near the conducting paper.

The experimental equipotential lines were nearly circular, consistent with the theoretical model for concentric electrodes. The minor distortions observed were likely caused by uneven electrode contact or imperfections in the conductive paper.

The Voltage vs Radius graph (Graph 3) confirmed the logarithmic dependence of potential on radius. The fit was close to theory across most of the range, validating Equation (16.6).

The Electric Field vs 1/r graph (Graph 4) supported the expected inverse relationship. The experimental points clustered around the theoretical line, though some scattered deviations occurred, particularly at small radii. These deviations likely arose from greater sensitivity to error near the inner electrode, where small radial measurement uncertainties cause larger field errors.

Overall, the results confirmed the theoretical predictions: potential varies logarithmically with radius, and the electric field decreases as 1/r. While small systematic errors were present, the experiment successfully demonstrated the relationship between potential and field in a coaxial geometry.

Conclusion

The primary goal of this experiment was to investigate the relationship between electric potential and electric field using two different electrode configurations: parallel plates and a concentric ring system. Both theoretical models and experimental data were compared to evaluate how closely physical measurements align with expected results.

In Investigation 1, the parallel plate setup produced equipotential lines that were nearly straight and evenly spaced in the central region, supporting the prediction of a uniform electric field. The measured electric field averaged – 0.95 V/cm, compared with the theoretical value of – 1.0 V/cm, with relative errors between 1% and 3%. Small deviations near the electrode edges highlighted the presence of fringe effects, were field lines curve outward, reducing uniformity.

In Investigation 2, the concentric electrode system generated equipotential lines that were approximately circular, as predicted. Voltage measurements decreased logarithmically with increasing radial distance from the central electrode, in agreement with the theoretical model. The experimental field values showed a general 1/r dependence, but with occasional scatter due to manual measurement uncertainty and electrode contact resistance. Despite these sources of error, the data trended closely toward the theoretical curves, demonstrating the expected relationship.

Overall, the experiment confirmed that:

- The potential difference directly influences field strength.
- Parallel electrodes create nearly uniform fields, while concentric electrodes create radially dependent fields.
- Experimental measurements align with theoretical predictions within a reasonable margin of error, validating both Gauss' law and the derived potential-field relationships.

Future improvements could include using more precise electrode contacts, reducing paper resistance variability, and employing digital data collection to minimize manual errors. These refinements would help further reduce uncertainty and bring measured values into even closer agreement with theory.

Questions

1. Doubling ΔV doubles the electric field since $E = \Delta V/d$ (Eq. 16.2). Equipotential lines compress closer together, and field line density increases.

2. Imperfections and poor contact introduce systematic errors, because they distort equipotential lines and shift measured values consistently across the sheet.

c. The average electric field in the fringe region is smaller than in the central region. Field lines curve outward and spread apart, reducing their density (Eq. 16.2).

d. Halving d while keeping ΔV constant doubles the field since $E \propto 1/d$ (Eq. 16.2).

e. For concentric electrodes, changing the outer radius affects the field logarithmically, according to

$$\mathrm{E}(r) = \frac{V_0}{(\ln b - \ln a)} \cdot \frac{1}{r}$$

(Eq. 16.5). Doubling the outer diameter changes the denominator slowly, so the field neither doubles nor halves, it changes, but not proportionally.

Acknowledgments

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